

Introduction

This project analyzes a publicly available dataset that represents a page-to-page graph of verified Facebook pages, providing a unique opportunity to explore the structure and interactions of mutual connections among official pages on one of the world's largest social media platforms.

Dataset Overview:

- **Nodes:** Each node represents a verified Facebook page.
- **Edges:** Edges denote mutual "like" relationships between two pages.
- **id_1:** Represents one verified Facebook page in a mutual connection.
- **id_2:** Corresponds to the other page that forms a mutual connection with id_1.

Each row in the dataset represents an undirected edge between two nodes. The dataset contains **22,470 nodes and 171,002 edges**, offering a rich network for analysis. Additionally, the dataset includes feature embeddings, which map numeric identifiers to corresponding lists of values. These embeddings are useful for implementing techniques like clustering and Principal Component Analysis (PCA) to uncover deeper insights.

Introduction & Dataset Description

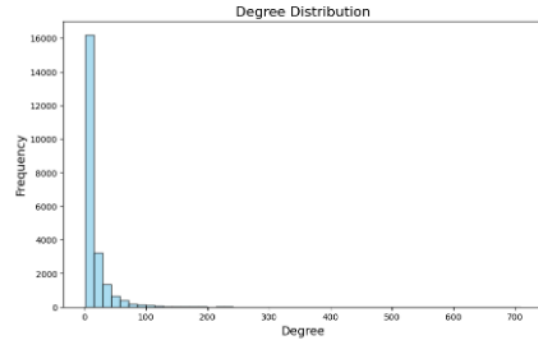
The analysis of social networks provides valuable insights into community interactions and influence distributions. This study examines a publicly available graph of verified Facebook pages, where nodes represent pages, and edges represent mutual likes. Our goal is to analyze the graph's structure and interactions to extract feature embeddings for machine learning tasks such as clustering and dimensionality reduction. By addressing key challenges, including computational complexity and low-connected nodes, we aim to provide a comprehensive understanding of the graph's properties and potential applications.

The dataset consists of 22,470 nodes and 171,002 edges. Each node represents a verified Facebook page, and each edge represents a mutual "like" relationship. The graph's fully connected nature ensures that all nodes are reachable from any other node, making it suitable for comprehensive analysis. The dataset also includes category labels for pages, enabling feature-category relationship analysis.

Exploratory Analysis

Key Graph Characteristics

- **Average Degree:** On average, each node is connected to 15.22 other nodes, indicating a moderately connected network.
- **Median Degree:** The median degree is 7, which suggests that a majority of nodes have relatively few connections, but a small fraction have many more.
- **Average Clustering Coefficient:** The average clustering coefficient of 0.3597 indicates that there is a moderate tendency for nodes to cluster together. This shows a reasonable degree of local connectedness within the network.
- **Connected Components:** The graph is fully connected, meaning there is only one connected component. All nodes are reachable from any other node.



Exploratory Data Analysis (EDA)

We performed an exploratory data analysis (EDA) to understand the graph's basic properties. The average node degree is 15.22, suggesting moderate connectivity, while the median degree of 7 indicates that most nodes have fewer connections. The average clustering coefficient of 0.36 shows that there is moderate local connectivity, and importantly, the graph is fully connected, meaning all nodes are reachable from any other node.

Key questions

1. Community Structure:
 - a. How many communities exist in the network, and what are their sizes?
 - b. What are the dominant categories within each community?
2. Influence Distribution:
 - a. Which page is the most influential within the network?
 - b. Which communities exhibit the highest overall influence?
3. Feature-Category Relationship:
 - a. Are there any significant differences in feature distributions across categories?

Key Questions

By framing the project around three key questions, we clarified the main focus areas, exploring Facebook pages' community structure, influence distribution, and feature relationships. These questions are significant as they not only unveil the overall network structure but also provide targeted guidance for business decision-making. Analyzing community structure offers insights into how nodes are organized within the network and the distribution patterns of different types of nodes. Influence distribution analysis further identifies key nodes and communities that play critical roles in the network, providing a basis for enterprises to formulate precise marketing and resource allocation strategies. Finally, studying the relationship between features and categories helps distinguish characteristics across different categories, offering data support for market segmentation and personalized marketing strategies. The analysis framework is progressive, with each question building on the previous, ultimately delivering a comprehensive view for business applications.

Community analysis

Questions:

1. How many communities can be identified?
2. What is the size of each community, and what are the dominant categories in each community?

Challenges:

- The graph is too large, making direct analysis computationally expensive.
- After sampling, a significant number of pages in the graph are disconnected.
- The normal Girvan Newman algorithm tends to produce a large number of single-node communities.

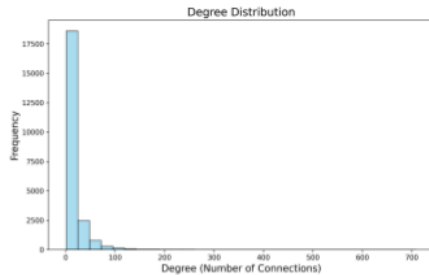
Approach:

- Degree-based sampling and cleaning method.
- Girvan Newman algorithm with minimum number of community nodes.
- Use silhouette scores as an additional evaluation measure.
- Multiple Minimum Community Threshold Experiments.

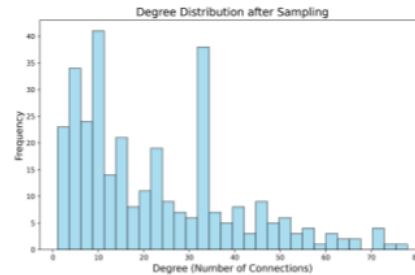
Community Detection Question

When analyzing the network's community structure, we faced three major challenges. First, the graph is extremely large, consisting of 22,470 nodes and 171,002 edges, which demands significant computational resources to analyze the entire structure. In fact, we initially attempted to analyze the full graph, but due to the excessive runtime, we had to explore alternative methods. Second, many nodes have low connectivity, which contributes minimally to the community structure analysis and can even distort results by affecting the division of highly connected nodes. Lastly, the traditional Girvan-Newman algorithm tends to produce single-node communities, which contradicts our goal of analyzing overall community characteristics. To address these challenges, we designed a degree-based sampling method, focusing on high-degree nodes to significantly reduce data size and improve analysis efficiency and effectiveness. Additionally, we modified the Girvan-Newman algorithm by introducing a node count threshold to avoid generating numerous insignificant small communities. We conducted experiments using thresholds of 10, 20, 30, 40, and 50 nodes per community. To ensure robust evaluation, we used both modularity and silhouette scores, as relying solely on modularity may introduce bias. After comparison, a threshold of 20 nodes produced the best overall results for modularity and silhouette scores. This method not only improved computational efficiency but also ensured the reliability and interpretability of the results.

Degree-based Sampling Method



- The number of nodes is very large.
- The complete data is mainly composed of low-connected pages.



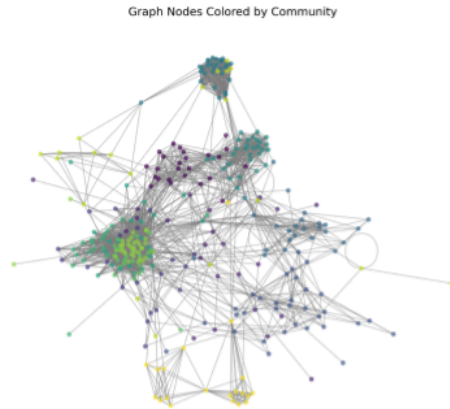
- Select the 1.5% node data with the highest number of edges in each category.
- Clean up communities with less than 20 nodes.
- 319 nodes and 3883 edges left.

Degree-based Sampling Method

To effectively address the issue of the graph's large scale, we adopted a degree-based sampling strategy during the data reduction phase, selecting the top 1.5% of nodes with the highest degree in each category. This method focuses on the key nodes that have the greatest impact on the network structure, significantly reducing data complexity while maintaining the original distribution of categories. We then set a minimum community size threshold of 20 nodes to filter out small, insignificant communities.

This process reduced the original dataset from 22,470 nodes and 171,002 edges to 319 nodes and 3,883 edges. It not only decreased the computational burden but also established a solid foundation for subsequent community detection. By preserving the category distribution in the original data and retaining the most critical node information, this sampling approach is particularly suitable for the structural analysis of large-scale networks.

Girvan Newman Algorithm



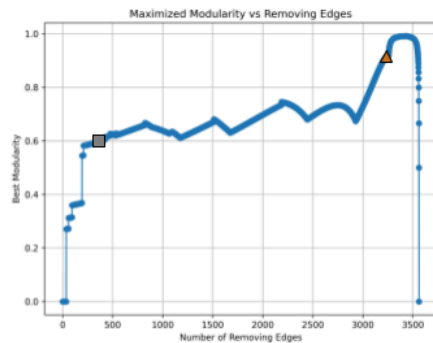
- Set 20 as the threshold of nodes in each community.
- Detect 13 communities in total.
- Government type of pages are the most dominant type in 5 of the communities.
- Company type of pages are the most dominant type in 3 of the communities.
- Politician and TVshow pages dominate in 2 communities each.

Modified Version of Girvan-Newman Algorithm

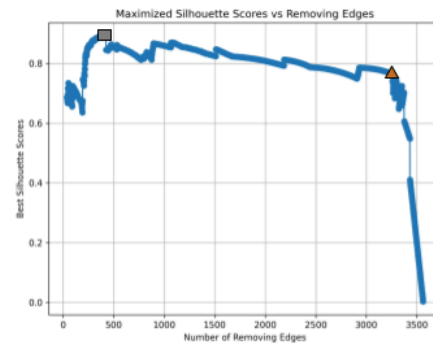
To enable the Girvan-Newman algorithm to detect communities with sufficient size and representativeness, we set a minimum threshold of 20 nodes per community. Under this restriction, modularity was used as a metric to identify the optimal community structure. Once no more edges could be removed, the restriction was lifted, but the best community structure was not updated further until all edges were removed.

Using this improved Girvan-Newman algorithm, we identified 13 significant communities within the network. Among these, five communities were dominated by government pages, highlighting their prominence; three communities were dominated by company pages, reflecting the importance of business nodes. Other types, such as politicians and TV shows, dominated two communities each. Additionally, the dominant category in each community accounted for more than 50% of its pages. This analysis not only reveals the concentrated distribution of categories within the network and the strong association between communities and categories but also provides critical insights into the interactions and distributions among communities. These findings offer a reliable foundation for further exploration of the relationship patterns and potential value within social networks.

Metrics



- The results show that when the minimum number of community nodes is set to 20, the optimal modularity (red triangle) is 0.92.
- When the minimum number restriction is lifted, the modularity value is still increasing.



- The best silhouette (green square) appears when only a small amount of edges are removed.
- The silhouette score plummeted after the minimum community restriction was lifted.

Metrics for Evaluation

To validate the effectiveness of community detection, we introduced two key metrics: modularity and silhouette scores. Modularity measures the compactness and rationality of community divisions, achieving its optimal value of 0.92, represented by the red triangle in the figure. This result indicates a well-structured community division. Silhouette scores reflect the separation between communities and the cohesion within them, achieving the best score when only a small number of edges were removed. Notably, the silhouette score remained relatively high throughout the process.

After removing the node count restriction, modularity initially increased slightly but then dropped rapidly, while the silhouette score plummeted to 0, indicating blurred community boundaries and significantly reduced quality of division. These results demonstrate that setting a reasonable node count threshold not only improves computational efficiency but also ensures the scientific validity and interpretability of community detection. The evaluation results provide strong evidence for the rationality of the algorithm design.

Analyzing influence among Facebook gages

Questions:

1. Which community demonstrates the greatest overall influence?
2. What are the most influential pages in each community?

Challenges Faced:

- The graph is very large and computationally expensive to process.
- Initial attempts at community detection and clustering took significant time and yielded unclear results.

Approach:

- Tried community detection on the entire graph, but had to quit (time-intensive).
- Used spectral clustering on full graph ($n_{clusters} = 5$).
- Performed random sampling of smaller graphs (2,000 and 10,000 nodes) for detailed analysis.
- Used community analysis and Spectral clustering.

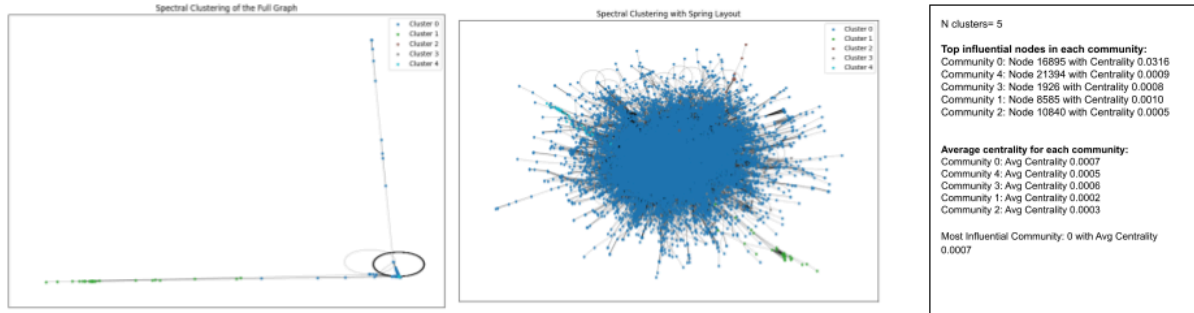
Influence Analysis

In this segment of the project, we focused on exploring the concept of influence within the Facebook graph network. The analysis was structured around three guiding questions:

1. Which community demonstrates the greatest overall influence?
2. What are the most influential pages within each community?

To tackle these questions, we attempted to assess influence across the graph's structure. However, the analysis was not without challenges. The graph's vast size posed significant computational hurdles. Processing the graph to detect communities and clusters required extensive time and resources, often leading to unclear or suboptimal results in our initial attempts. This necessitated iterative refinement of our methods to manage the computational complexity effectively and derive meaningful insights.

Spectral clustering on full-graph



Drawbacks:

- Insufficient Number of Clusters: $n=5$ may be too small, as the data appears to contain more clusters.
- Cluttered Visualization: The visual representation is too cluttered, making it difficult to distinguish the individual

Next steps:

- Elbow plot computation for the entire graph took too long as we increased n beyond 5.
- We decided to create a subgraph of 10,000 nodes and another with 2000 nodes to improve performance and better analyze the clusters.

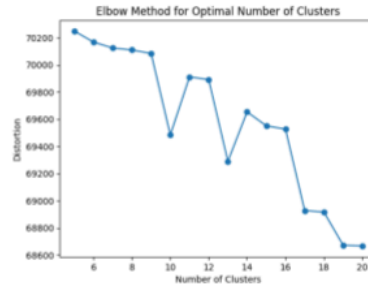
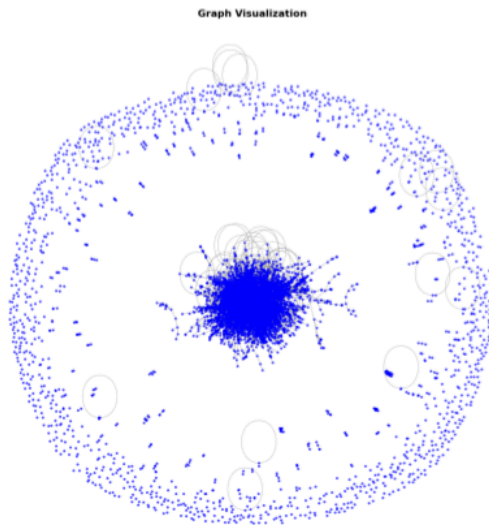
Spectral Analysis

Our initial approach involved performing spectral clustering on the full graph, starting with five clusters. This method provided some preliminary insights into the community structure.

However, we quickly recognized that using only five clusters was insufficient to capture the graph's complexity. The resulting visualization was overly cluttered, making it difficult to discern clear patterns or interpret the data meaningfully.

To refine our approach, we attempted to generate an elbow plot for a higher number of clusters. However, this process proved computationally expensive and time-consuming due to the graph's size. Consequently, we shifted our strategy towards creating subgraphs, allowing us to improve computational performance and analyze the clusters more effectively.

Random sample of 10k nodes



We created a sub-graph by randomly sampling 10000 nodes to explore a larger portion of the graph and analyze its structure with spectral clustering.

Key Characteristics of the Sub-Graph:

- **Nodes and Edges:** 10000 nodes and 35,945 edges.
- **Connected Components:** 1327 connected components, with the largest one containing 8285 nodes (83% of the total graph).
- **Average Clustering Coefficient:** 0.3306

Elbow plot: We used an elbow plot to determine the optimal number of communities, finding that $n = 10$ was the best choice for community detection.

Subgraph Creation and Cluster Analysis

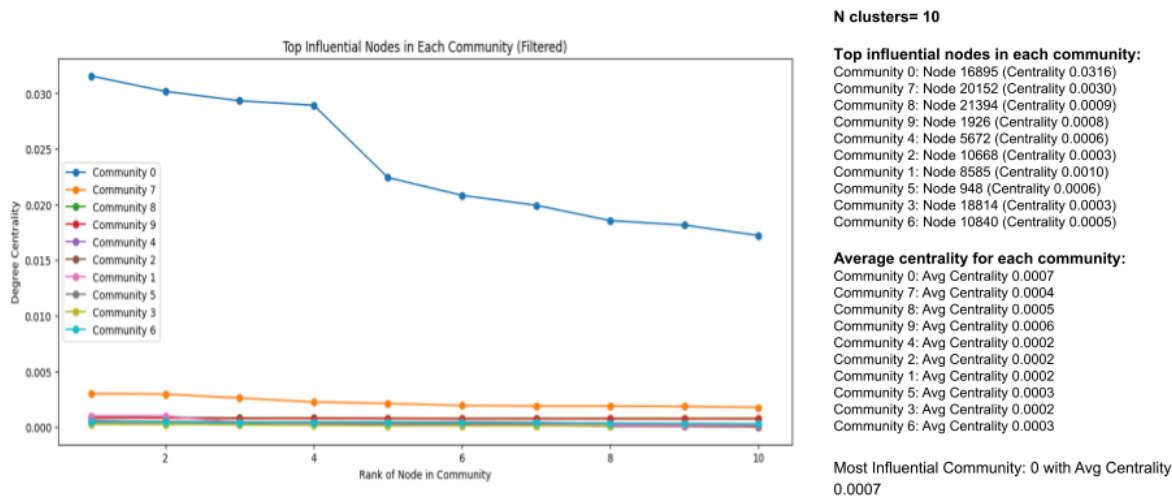
To address computational challenges, we created a subgraph by sampling 10,000 nodes from the original graph. This subgraph contained 35,945 edges and 1,327 connected components. While random sampling introduced some disconnected nodes, the subgraph retained enough data to provide meaningful insights while significantly reducing computation time.

Key observations included:

- The largest connected component within the subgraph comprised 8,285 nodes.
- The average clustering coefficient for the subgraph increased to 0.33, reflecting stronger local connectivity compared to the full graph.

We then generated an elbow graph to determine the optimal number of clusters, testing values from 6 to 20. The plot revealed a significant drop in distortion values between 10 and 11. Based on this result, we selected the optimal number of clusters for further analysis.

Spectral clustering on 10k nodes sub-graph



Spectral Clustering Results

Using spectral clustering on the largest connected component of the subgraph, we successfully identified 10 clusters. Among these, Community 0 emerged as the most influential, with an average centrality score of 0.0007. The most influential node within Community 0 exhibited a centrality of 0.0316.

The analysis revealed significant variations in centrality scores across the identified communities. Some clusters exhibited very low centrality, indicating minimal influence, while others, like Community 0, demonstrated high centrality values, signifying greater influence. This variation underscores the heterogeneous nature of influence within the Facebook graph network and highlights the importance of tailored community-specific strategies in network analysis.

Feature-Category Relationship

Questions:

1. To examine the relationship between features and categories, are there noticeable differences in the distribution of pages across categories in the feature space?

Challenges Faced:

- The exact nature and description of these embeddings (e.g., features, connections or some latent representation) remains unspecified from the data.

Approach:

- Tried dimensionality reduction by PCA and t-SNE.
- Cluster Analysis through K-Means Clustering.
- Statistical Analysis of feature variations..

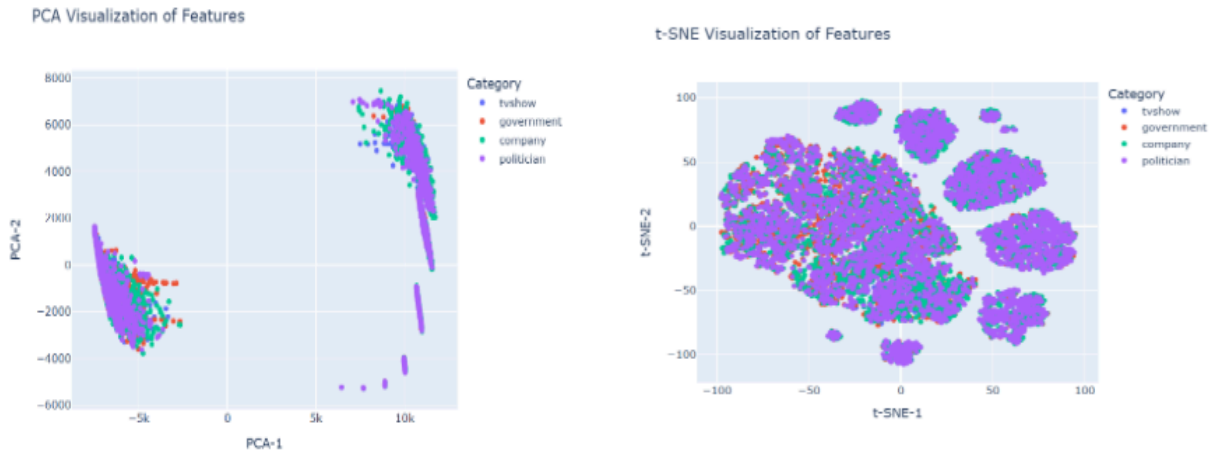
Feature-Category Relationship

We examined feature-category analysis by addressing one key question:

1. To examine the relationship between features and categories, are there noticeable differences in the distribution of pages across categories in the feature space?

The main challenge in this analysis lies in the lack of clarity regarding the exact nature and interpretation of these embeddings, which could be specific features, relationships, or latent representations they capture from the data.

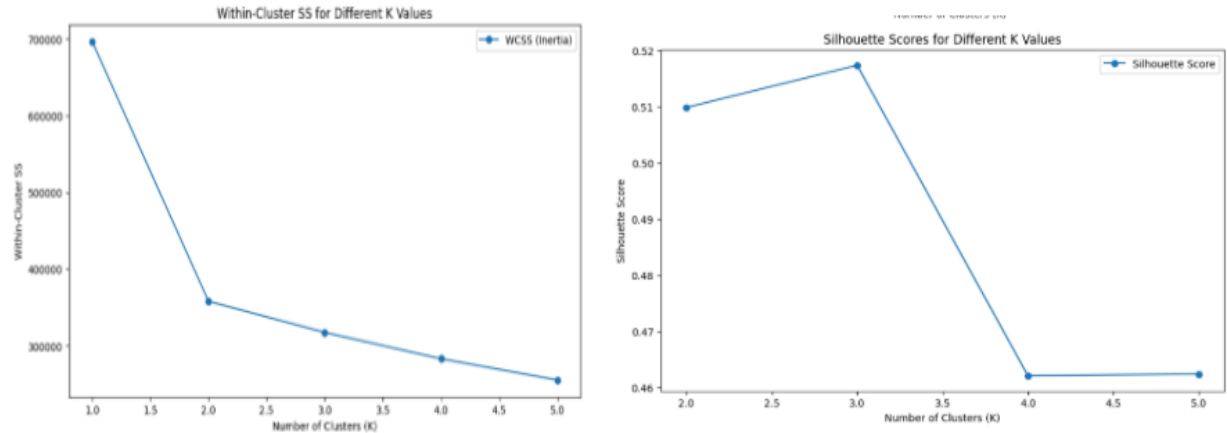
Dimensionality Reduction by PCA and t-SNE



Dimensionality Reduction by PCA and t-SNE

We focussed on implementing PCA and t-SNE plots with a number of components equal to 2. From the PCA visualization, it appears that the clusters of data points overlap significantly across categories. This is because PCA may fail to capture non-linear relationships in the data, leading to overlapping clusters when the features have complex dependencies. The t-SNE plot shows tighter and more distinct clusters compared to PCA. From the t-SNE plot, the purple category (Politician) seems to form tightest clusters. This suggests that its features are well-represented and distinct, with less overlap with other categories. Other categories, such as red category (government) and green category (company), seem more dispersed and intermixed and overlap significantly in the feature space.

Cluster Analysis through K-Means Clustering



Cluster Analysis through K-Means Clustering

We also implement K-Means clustering with the number of clusters equal to 5 and max number of iterations equal to 100. We compute within cluster sum of squares and silhouette scores for each iteration and also implement their plots.

The "Within-Cluster SS (WCSS)" plot is used to identify the optimal number of clusters (K). Through the elbow method, we see a sharp decline in WCSS between K=1 and K=2, followed by a more gradual reduction, indicating that three clusters may effectively capture the structure of the data. This suggests meaningful groupings in the feature space and also potentially representing distinct category patterns.

Similarly, the "Silhouette" plot evaluates the quality of clustering. From the Silhouette plot, we find that K=3 achieved the highest score, indicating well-defined clusters. Beyond K=3, the drop in the score suggests that adding more clusters results in overlapping or less distinct groupings.

Statistical Analysis of feature variations

```
ANOVA Results: {0: {'F-statistic': 0.5785699301136863, 'p-value': 0.6290303625976599}, 1:
{'F-statistic': 0.6905485697018646, 'p-value': 0.5576803627318372}, 2: {'F-statistic':
0.14591469895768638, 'p-value': 0.9323385185362577}, 3: {'F-statistic': 173.93740616278492, 'p-value':
1.7472573908538236e-111}}
```

```
T-test Results: {0: {'T-statistic': 0.9864386796810508, 'p-value': 0.3239412789302136}, 1:
{'T-statistic': -1.1201536442356543, 'p-value': 0.262674655300398}, 2: {'T-statistic':
0.2932786904360751, 'p-value': 0.7693151039281285}}
```

Statistical Analysis of feature variations

We perform ANOVA and T-tests between two categories: TV-Show and Government.

From the ANOVA test, features with high F-statistics and low p-value indicates significant differences in the distribution of the feature across categories such as Feature 3, which has a F-statistic of 173.93 and p-value of $1.74 \cdot 10^{(-111)}$. From the T-tests, features with high T-statistics and low p-value indicate a large difference between two groups for a given feature. From the above figure, we have no features which have high T-statistic and low p-value.

Conclusion

Recommendations to the business

1. **Target High-Centrality Nodes:** Focus on **Community 0** and its key node (Node 16895) for marketing, communication, or influencer engagement strategies, as it holds the highest potential for reaching a broader audience.
2. **Niche Strategies for Smaller Communities:** Communities with lower centrality scores could be targeted for niche or specialized interventions, helping build influence within those groups over time.
3. **Monitor Community Dynamics:** Continuously monitor the centrality scores of nodes in smaller communities, as changes in centrality over time could indicate emerging influencers or shifts in network dynamics.
4. **Tailored Engagement:** Based on the identified communities, customize outreach and engagement strategies to either focus on influential nodes for rapid growth or address the needs of lower centrality communities for long-term engagement.
5. **Tailored Business Strategies:** For overlapping categories, adopt more flexible and generalized strategies that account for the possibility of shared traits. For example, develop hybrid offerings or messaging that appeals to both government and company stakeholders.

Conclusion

Based on our analysis, we recommend the following business strategies:

Target High-Centrality Nodes: Focus on Community 0 and its key node (Node 16895) for marketing and influencer engagement, as it has the highest potential to reach a broad audience.

Niche Strategies for Smaller Communities: Communities with lower centrality scores could be targeted for specialized interventions to gradually build influence.

Monitor Community Dynamics: Track changes in centrality scores to identify emerging influencers and shifts in community influence.

Tailored Engagement: Customize outreach strategies for each community based on their centrality and influence levels to optimize engagement.

This study demonstrates the power of graph analysis in uncovering community structures, influence distributions, and feature relationships. By addressing computational challenges and leveraging advanced techniques, we provide actionable insights that can inform practical applications in social network analysis.